

Hrishikesh Pawar

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Education

Worcester Polytechnic Institute

August 2023 - May 2025 (Expected)

Master of Science in Robotics Engineering; GPA: 3.88/4.0

Worcester, MA

Courses: CS541-Deep Learning, RBE549-Computer Vision, RBE595-Vision based Robot Manipulation, REB550-Motion Planning

Teaching Assistant: RBE595-Vision based Robot Manipulation, RBEBL01-Robots for Recycling

Experience

PeAR Group | Graduate Researcher | Worcester, MA

August 2024 – Present

- Built a **Depth-from-Defocus** navigation framework using **event cameras**, achieving **62x** computational efficiency over **RGB-D** baselines
- Created a custom **3D Gaussian Splats** simulator rendering at **100 FPS** for real-time **HITL** testing of event-based vision systems
- Investigate low-light drone navigation using **structured lighting** and **physics-informed** neural networks to address the **sim-to-real** gap

Nobi Smart Lights | AI Software Intern | Remote

May 2024 - August 2024

- Led R&D efforts on ceiling-mounted, vision-enabled lamps supporting real-time **fall detection** and emergency response in elderly care
- Engineered a **rotation-aware** detection model by integrating **Swin Transformers**, improving accuracy from **85 mAP** to **92 mAP**
- Explored **LLaVA** and **CLIP** with **LoRA-based PEFT** for fall-context inference, evaluated on **100+** scene-prompt pairs with **61%** baseline score
- Automated deployment pipeline using **Jenkins**, **Docker**, and **Kubernetes**, reducing rollout time by **30%** for new ML models on edge devices

Adagrad AI | Computer Vision Engineer | Pune, India

November 2020 - July 2023

- Developed core modules for **Gate-Guard**, an **edge-based** system for **vehicle access control** using **Automatic License Plate Recognition**
- Achieved **92%** accuracy for **four-wheelers** and **88%** for **two-wheelers** at **50 FPS** on **NVIDIA Jetson TX2**, with deployment across **100+** sites
- Built real-time dashboards with **Django** and **Azure**, using backend services (**Kafka**, **Celery**, **Redis**, **WebSockets**) to handle **3K+ scans/hour**
- Featured links: [Indian Express](#) | [Times of India](#) | [YouTube](#)

Publications

Hrishikesh Pawar, Deepak Singh, Nitin J Sanket; Proceedings of the Winter Conference on Applications of Computer Vision (WACV) Workshops, 2025, pp. 912-916 | [Publication](#)

- Proposed a bio-inspired navigation strategy that leverages **optical defocus** in **event cameras** for energy-efficient aerial perception
- Modeled passive **sharpness-based** depth cues from defocus patterns to **segment foreground obstacles** from **background regions**
- Achieved **82%** navigation success rate with **62x** computational savings compared to state-of-the-art methods (**Depth-Pro**, **MiDaS**)

Projects

Deep Visual-Inertial Odometry | [GitHub](#) | [Project Report](#)

- Designed a **VIO** pipeline coupling **CNNs** for visual features and **LSTMs** for **IMU** integration to improve pose estimation accuracy
- Simulated motion trajectories across **3** noise profiles and **25** trajectory types in **Blender** to build synthetic training data
- Optimized training using **Mean Squared Error** and **Geodesic loss**, resulting in a **28%** reduction in **RMSE** of absolute trajectory error

Zero-Shot Semantic Neural Style Transfer for Images | [GitHub](#)

- Implemented **AdaAttN** module to perform per-point adaptive normalization for **zero-shot semantic neural style transfer** tasks
- Achieved a **13%** training time reduction by removing redundant deep attention layers while maintaining visual style fidelity
- Integrated **AdaAttN** with **CLIPSeg's** prompt-based segmentation for region-specific style transfer across **50+** semantic prompts

Skills

- **Languages:** Python, C, C++, SQL, Bash, MATLAB
- **Frameworks:** PyTorch, OpenCV, TensorFlow, ONNX, NumPy, Blender, SciPy, Scikit-learn, Matplotlib, Pandas, Django, Flask, Celery, Kafka, CMake, Docker, TensorRT, Jenkins, Kubernetes, Git, ROS, ROS2

Leadership and Achievements

- **Presentation:** Presented research on defocus-based depth estimation with event cameras at [WACV 2025 Workshop](#) — [Link](#)
- **Recipient:** [Dr. Glenn Yee Graduate Student Project Award](#) — [Link](#)
- **Recognition:** Won **Best Project Award** in [WPI's RBE-549 Computer Vision](#) course, selected from a cohort of **40** graduate students — [Link](#)
- **Leadership:** Led **56-member** team in [Formula Student competitions](#); **rank-1** at [Formula Bharat](#) and **top-30** at [Formula Student Germany](#)